Name	Akshara Balreddygari
Student ID	34975551
Studio Session	11

Part A: URL of the Visualisation

https://ak-ak-ak.github.io/FIT3179-DV2-FruitBowl/

Part B: Domain Description

Describe your domain chosen for DV II

My visualisation explores Australian fruit production through the question, “Where does your fruit bowl come from?” It looks at which fruits make up Australia’s harvest, where they grow, how production and value have changed, and where selected fruit exports go.

The main production data comes from the ABS Australian Agriculture: Horticulture release for 2024–25, with historical comparisons covering 2020–21 to 2024–25. I also use Geoscience Australia’s state land areas to calculate production density and UN Comtrade data, accessed through World Bank WITS, for exports in calendar 2025. Natural Earth provides supporting map boundaries.

Describe the Why and Who aspects of your DV II

I designed the visualisation for everyday Australians who may recognise the fruit in their supermarket but know less about where it is grown. Readers do not need agricultural knowledge to follow the story. The headings introduce the main findings, annotations explain important comparisons, and tooltips provide exact values for readers who want more detail.

The aim is to help readers compare the size and value of different fruit industries, recognise how strongly some crops depend on particular states, and understand that increases in production do not always correspond to increases in value.

In Munzner's task abstraction, the page supports identifying extremes, comparing quantities and proportions, and summarising spatial patterns and trends. It is primarily a presentation rather than an open-ended exploration. The four sections guide readers through the story, while the controls let them inspect particular fruits without losing the surrounding context.

What and How for Each Chart

1. Waffle Chart

What: This chart uses ABS production data for 29 fruit categories in 2024-25. It shows the five largest categories by weight, with the remaining categories grouped as "Other reported fruit".

How: I chose a waffle chart to make production shares easy to understand as parts of 100. The marks are equal-sized rectangles. The channels are count, which represents rounded percentage share, and colour hue, which identifies the fruit group. Oranges take up 20 boxes, making their share clear. Each box represents one percentage point of production weight, not one fruit. Tooltips show the exact shares.

2. Lollipop Chart

What: This chart shows the ten fruit categories with the highest farmgate values in 2024-25. Table grapes lead, followed by bananas and avocados. The nearby $7.33 billion total covers all 29 categories.

How: I chose a lollipop chart so readers can compare values on the same scale, particularly close values such as bananas and avocados. The marks are dots and horizontal rules. The channels are horizontal position and rule length for farmgate value, vertical position for fruit category, and colour hue for fruit identity. Sorting from highest to lowest makes the ranking easy to follow without implying a hierarchy.

3. Heatmap

What: This chart compares ten fruits across Australia's states and territories. Each reported cell shows the state's share of that fruit's national production in 2024-25.

How: I used a heatmap so readers can spot fruits that depend heavily on one state. The marks are rectangular cells. The channels are row position for fruit, column position for state, and sequential colour lightness for production share. Darker cells show larger shares, while grey means not reported. Outlines highlight Queensland's bananas, NSW's blueberries and Victoria's pears. Exact percentages stay in tooltips to avoid crowding the cells.

4. Choropleth Map

What: This map combines ABS production data with state land areas from Geoscience Australia. It shows tonnes produced per square kilometre of total state land for the selected fruit or all reported fruit.

How: I chose a choropleth to compare production density geographically rather than map raw totals. The marks are geographic areas representing states and territories. The channels are geographic position for location and sequential colour lightness for density, with darker blue showing higher values. Grey means not reported. A conic equal-area projection preserves geographic area relationships, and the annotation names the state with the highest reported density. This measure is not crop yield.

5. Proportional Symbol Map

What: This map uses experimental ABS regional production estimates for bananas and avocados in 2024-25. These fruits have the detailed regional data needed for this comparison.

How: I used proportional symbols to compare regional totals without giving physically larger regions more visual importance. The data marks are circles. The channels are geographic position for regional location, circle area for production tonnes, and colour hue for the selected fruit. Both fruits share the same area scale. Close-ups make North Queensland and south-west WA easier to inspect without moving the circles. The side list gives exact totals for leading regions nationally.

6. Sankey Diagram

What: This chart connects states and territories to the ten comparison fruits using positive reported production observations from 2024-25. The links total about 2.089 million tonnes.

How: I chose a Sankey so readers can follow both a state's harvest composition and the states contributing to each fruit. The marks are rectangular nodes and curved links. The channels are link width for production tonnes, node height for the total attached to each

node, and colour hue for fruit identity. Horizontal position separates states from fruits. Opacity highlights the selected state without changing quantities or layout. These links show production relationships, not transport.

7. Slope Chart

What: This chart compares production for ten fruits in 2020–21 and 2024–25. Selected labels show percentage changes, including avocados increasing by about 68% and pears falling by about 42%.

How: I chose a slope chart to make increases and decreases between the two years easy to spot. The marks are points and connecting lines. The channels are horizontal position for year, vertical position for production tonnes, colour hue for fruit identity, and opacity for highlighting. Lines connect each fruit's two observations. Percentage labels help readers recognise large relative changes as well as changes in tonnes. The lines do not show the intervening annual trend.

8. Bump Chart

What: This chart ranks ten fruits by farmgate value each year from 2020-21 to 2024-25. Table grapes remain first, while bananas and avocados finish above apples.

How: I chose a bump chart to show changes in ranking separately from dollar amounts. The marks are points and connecting lines. The channels are horizontal position for financial year, vertical position for rank, colour hue for fruit identity, and opacity for highlighting. Rank 1 sits at the top, and crossings show when fruits swap places. Endpoint labels and the summary help readers follow the rankings. Rank gaps do not represent dollar gaps.

9. Connected Scatterplot

What: This chart compares production tonnes with nominal farmgate value for a selected fruit across five financial years. Readers can choose from the ten comparison fruits.

How: I chose this chart to show whether production and value changed together. The marks are points and connecting lines. The channels are horizontal position for production tonnes, vertical position for farmgate value, and colour hue for the selected fruit. Lines connect points in financial-year order, with text labels identifying each year. The summary reports endpoint percentage changes. Both axes adjust to the selected fruit, so positions should not be compared directly across selections.

10. Waterfall Chart

What: This chart breaks down the $1.486 billion increase in reported fruit value between 2020-21 and 2024-25. It shows the five largest gains, two largest declines and the combined change in the other 22 categories.

How: I chose a waterfall to explain how the contributions add up to the overall increase. The marks are horizontal rectangles and connector rules. The channels are bar length for contribution size, horizontal position for the cumulative start and end values, vertical position for category, and colour hue for gains, declines and the total. Signed labels give exact changes. Table grapes and avocados make the largest individual gains, while olives and pears reduce the increase.

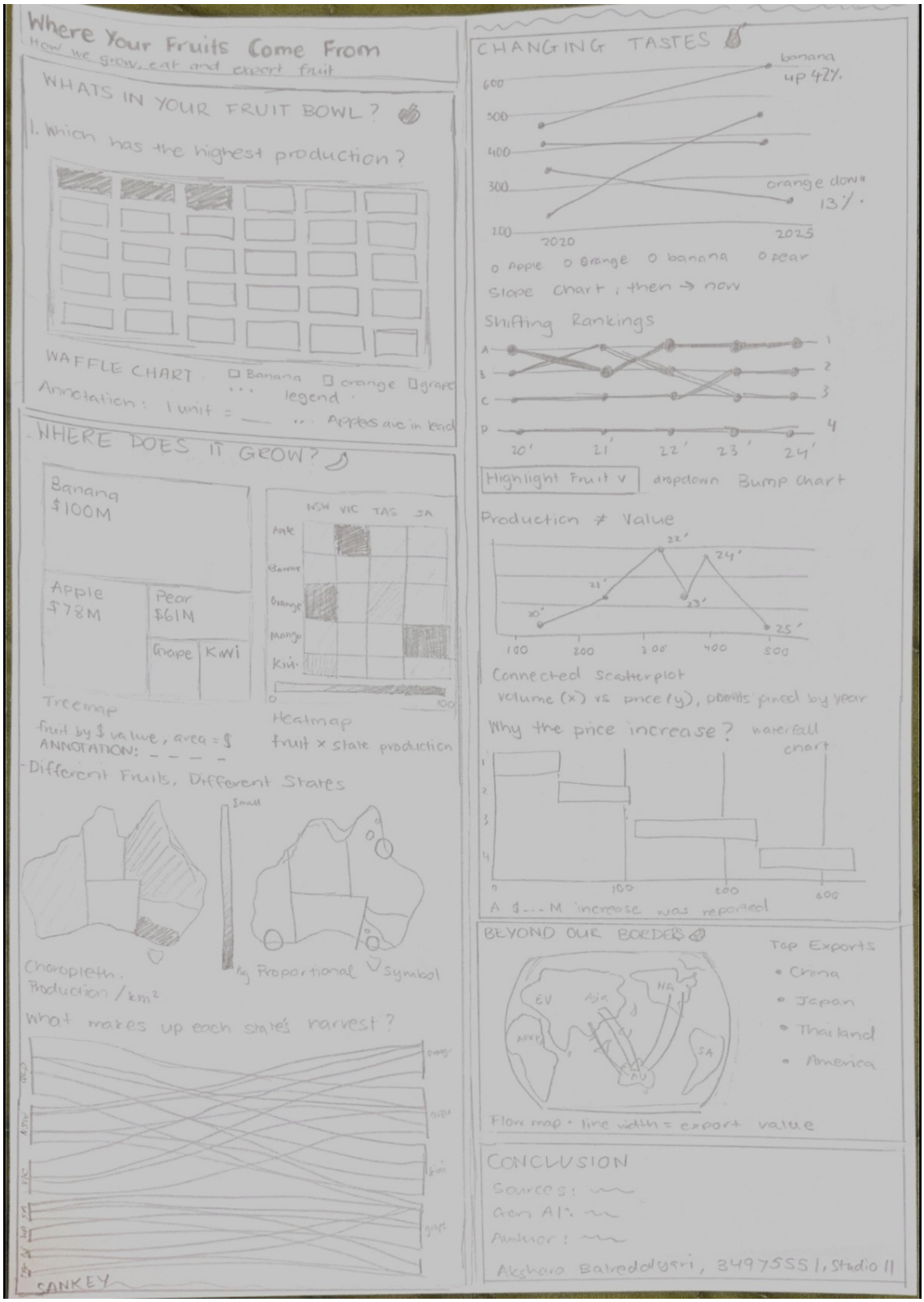
11. Export Flow Map

What: This map shows the eight largest export destinations for five selected fruit categories in calendar 2025. Shares use the full export total for the selection, including destinations not drawn.

How: I chose a flow map to show where the overseas markets are while comparing export volumes. The marks are geographic areas, connecting lines and point symbols. The channels are geographic position for location, line width for export tonnes, and colour hue for destination identity. Colours match the accompanying table. Point size is fixed and does not encode quantity. The map uses an Equal Earth projection, and line widths rescale for each selection, so comparisons should stay within that selection. The lines are not actual shipping routes.

Part C: Sketch

https://github.com/ak-ak-ak/FIT3179-DV2-FruitBowl/blob/main/docs/FIT3179%20DV2%20Sketch.pdf



Part D: Full-size screenshot of your DV II

AUSTRALIAN FRUIT PRODUCTION · 2024–25

Where Your Fruit Bowl Comes From

Where Australia grows its fruit, how production is changing, and what reaches overseas markets.

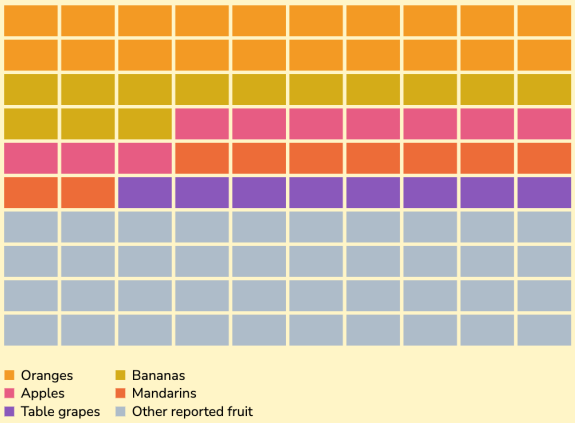


What's in the bowl?

Australia grows a wide variety of fruit, but production is concentrated in particular regions. Follow where it grows, how production is changing, and where some of it goes overseas.

Oranges take the biggest share by weight

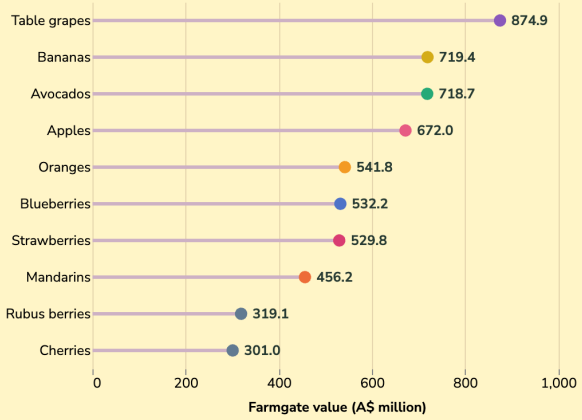
Oranges make up 20% of the 2.835 million tonnes reported in 2024–25.



One unit is one percentage point of reported production weight, not one fruit or serving. Rounded units total 100; tooltips show exact shares. Source: ABS, Table 2.

Table grapes lead by value

The **ten highest-value fruits** in 2024–25. Farmgate value is the value of fruit when it leaves the farm, before retail costs.



Across **all 29 reported categories**, farmgate value totals **$7.33 billion**, not supermarket prices. Wine grapes are excluded. Source: ABS, Table 2.

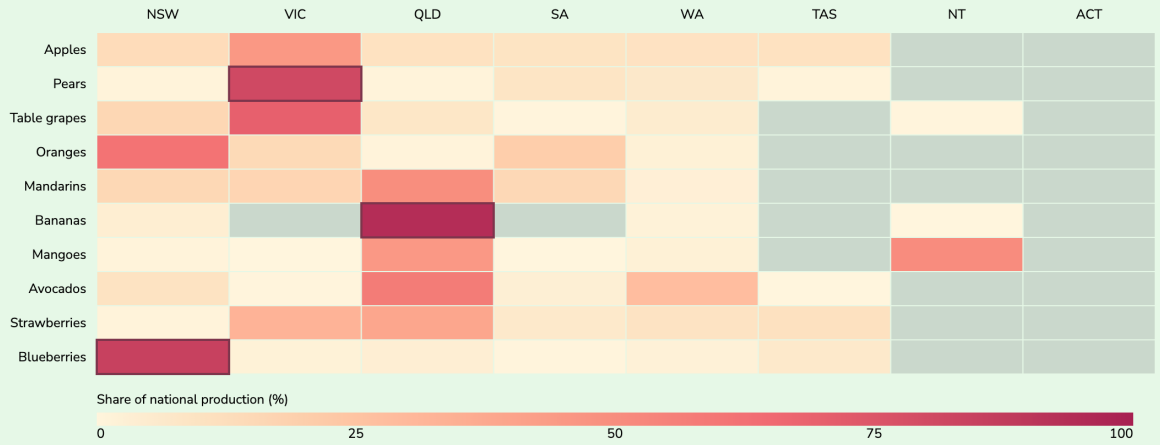
▶ Data notes

Where it grows

Different fruits come from different parts of Australia. Compare the states first, then look more closely at the growing regions of bananas and avocados.

Some fruits depend heavily on one state

94% of bananas come from Queensland, 85% of blueberries from NSW, and 81% of pears from Victoria.



Colour shows each state's share of that fruit's national production. Grey means not reported, not zero; hover for exact percentages.
Source: ABS, Table 2, 2024–25.

▶ State names and data notes

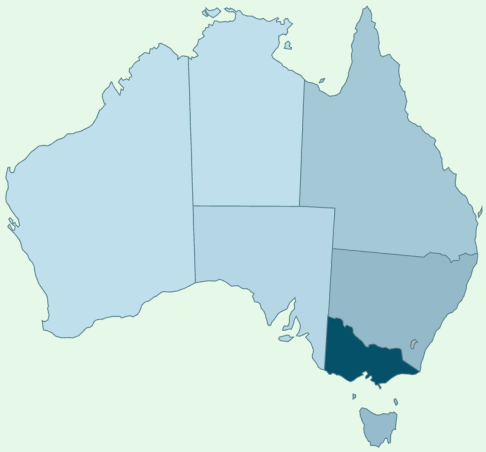
Different fruits, different growing states

Reported production per square kilometre of state land, 2024–25.

Fruit

- (●) **All reported fruit**
- ( ) Apples
- ( ) Avocados
- ( ) Bananas
- ( ) Blueberries
- ( ) Mandarins
- ( ) Mangoes
- ( ) Oranges
- ( ) Pears
- ( ) Strawberries
- ( ) Table grapes

Highest reported density
Victoria
3.042 tonnes per km² of state land





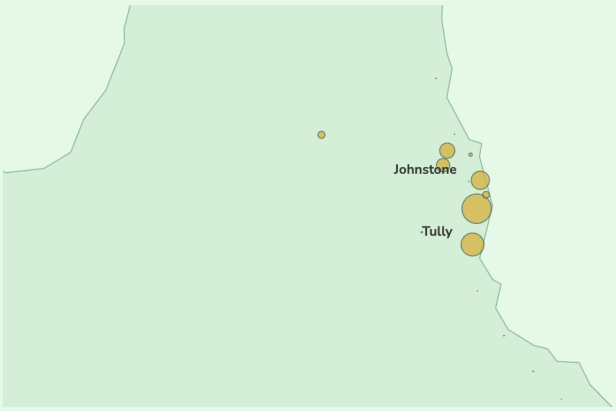
All reported fruit: Victoria has the highest reported density, 3.042 tonnes per km² of total state land. This is not orchard yield.

Grey means not reported. Density uses total state land, not orchard area or yield. Sources: ABS and Geoscience Australia.

▶ Data notes

Zooming in on the growing regions

Bananas cluster in **tropical North Queensland**; avocados have another major production centre in **south-west WA**.



Bananas: leading regions nationally

1. **Johnstone**: 124,529 tonnes
2. **Tully**: 75,607 tonnes
3. **Babinda**: 48,238 tonnes
4. **Mareeba**: 33,230 tonnes

Names and exact totals help separate the nearby growing regions. Circle area uses the same scale in every view.

Production (tonnes)

1,000 20,000 100,000

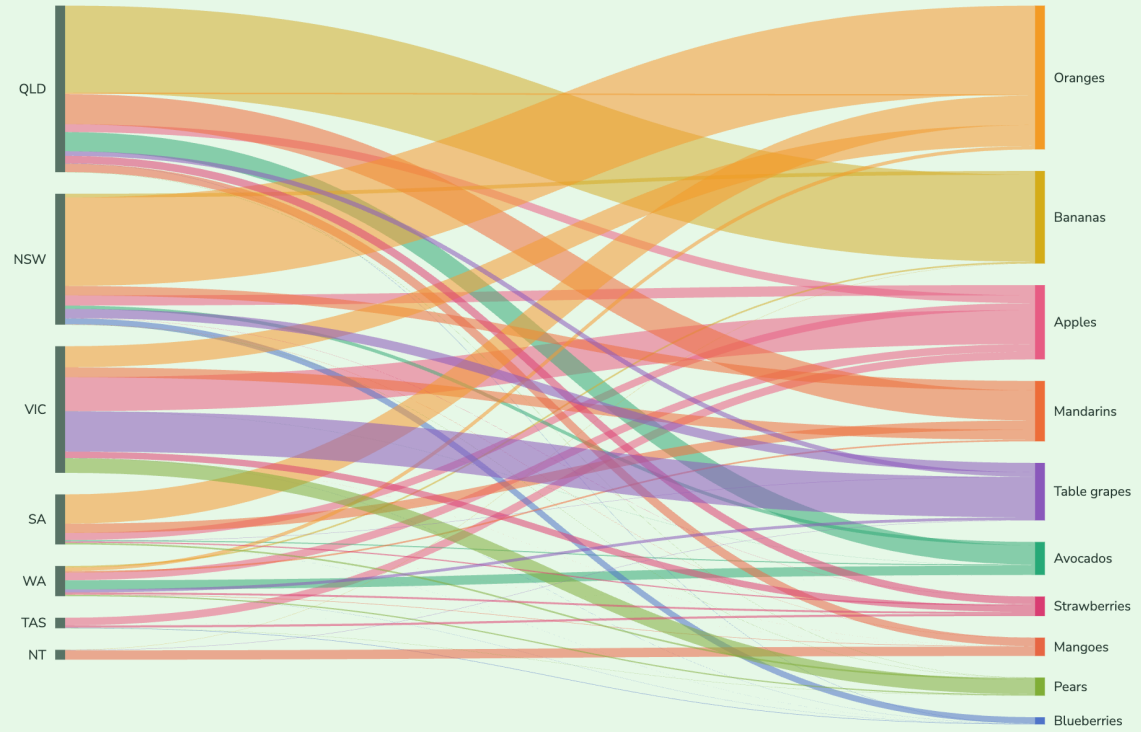
Fruit: Bananas Region: North Queensland

Circle area represents tonnes on a shared scale for both fruits. Points locate regions, not farms. Source: ABS, experimental 2024–25 estimates.

▶ Regional coverage

What makes up each state's harvest?

Queensland's banana links and NSW's orange links stand out. Link widths compare reported tonnes across ten fruits in 2024–25.



Highlight state: All states

Link widths show **2.089 million reported tonnes** across ten fruits, not transport. Missing observations are not zero. Source: ABS, Table 2.

▼ Coverage and layout

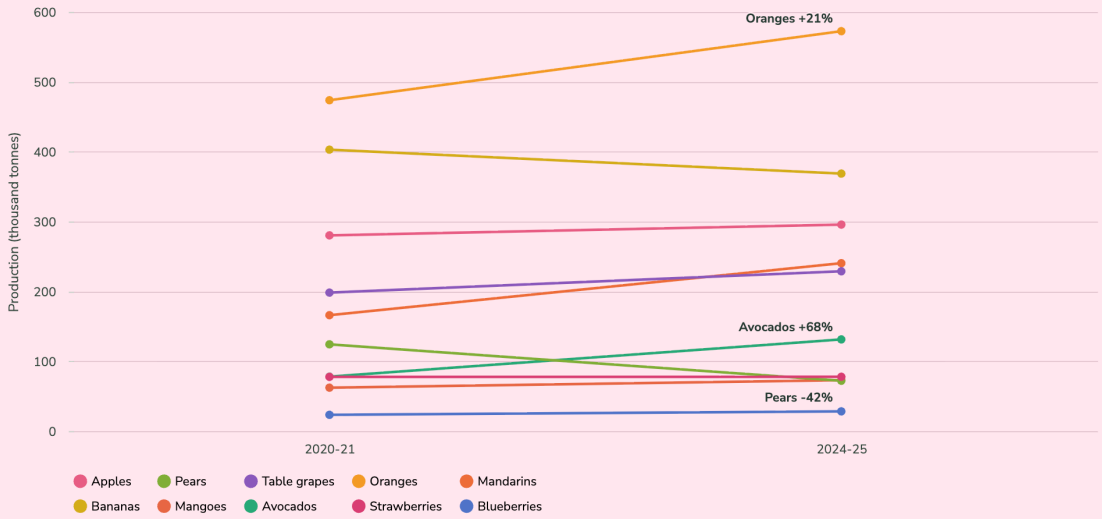
Only positive reported observations are linked. ACT has no positive reported observations in this comparison. d3-sankey calculates the layout during preparation; Vega draws the diagram. Highlighting a state retains all node positions and the original scale.

How it has changed

Fruit production has changed unevenly. Some crops have grown substantially, while others have declined. Production volume tells one part of the story; the value of that fruit tells another.

More avocados and mandarins, fewer pears

Between 2020–21 and 2024–25: **avocados +68%**, **mandarins +45%**, and **pears −42%**.

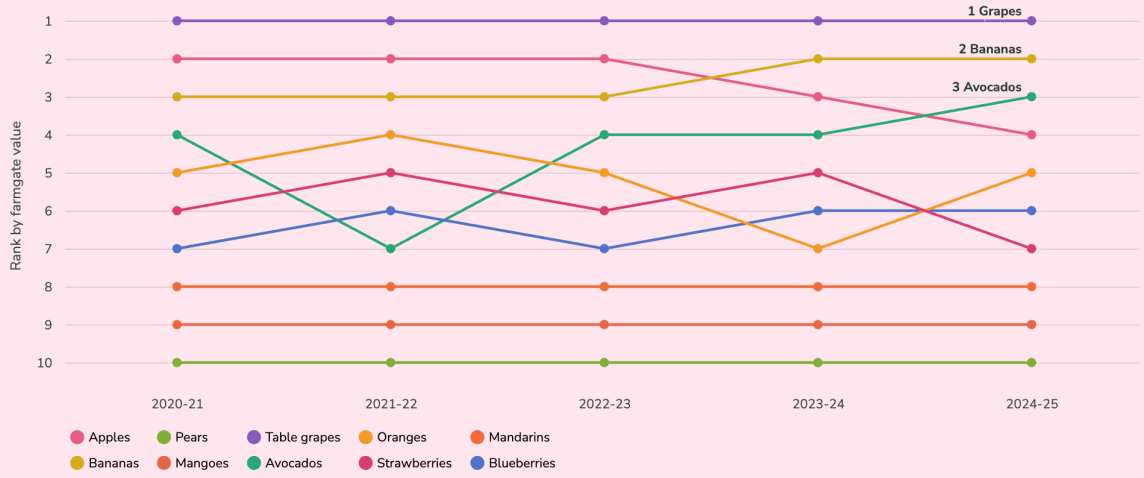


Highlight fruit: All fruits

Lines join two snapshots, not the intervening annual trend. The vertical axis shows thousands of tonnes and starts at zero. Source: ABS, Table 2. Percent changes calculated from published tonnes.

The value rankings have shifted

Table grapes stayed first among these ten fruits; bananas and avocados both finished above apples.



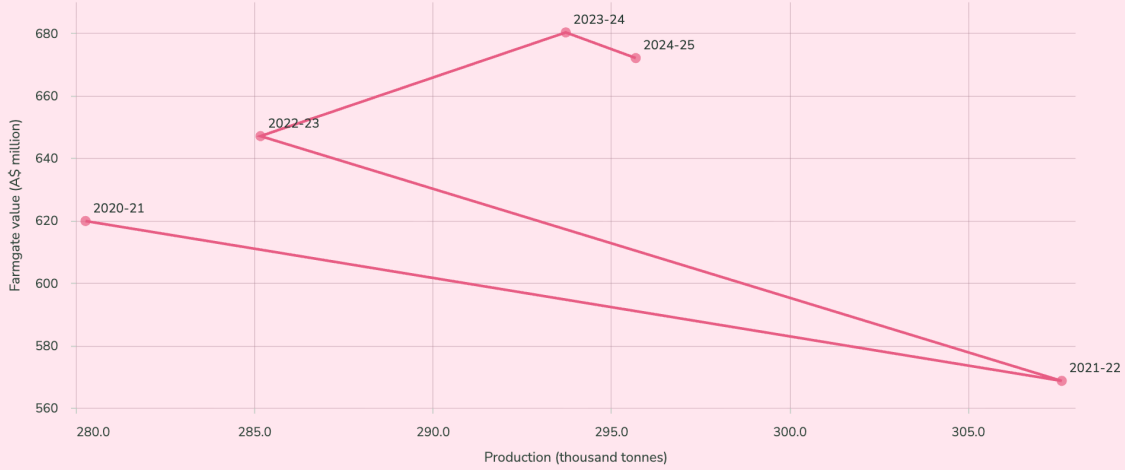
Highlight fruit: All fruits

> Avocados: rank 4 in 2020–21 and rank 3 in 2024–25 among these ten fruits. A rise in rank describes order, not the size of the value gap.

Rank 1 is the highest farmgate value within this fixed group of ten fruits. The chart shows changes in order, not the size of the dollar gaps. Values are in current dollars. Source: ABS, Table 2, 2020–21 to 2024–25.

More fruit does not always mean more value

Production volume and farmgate value do not always move together. Each labelled point is one financial year.



Fruit: Apples

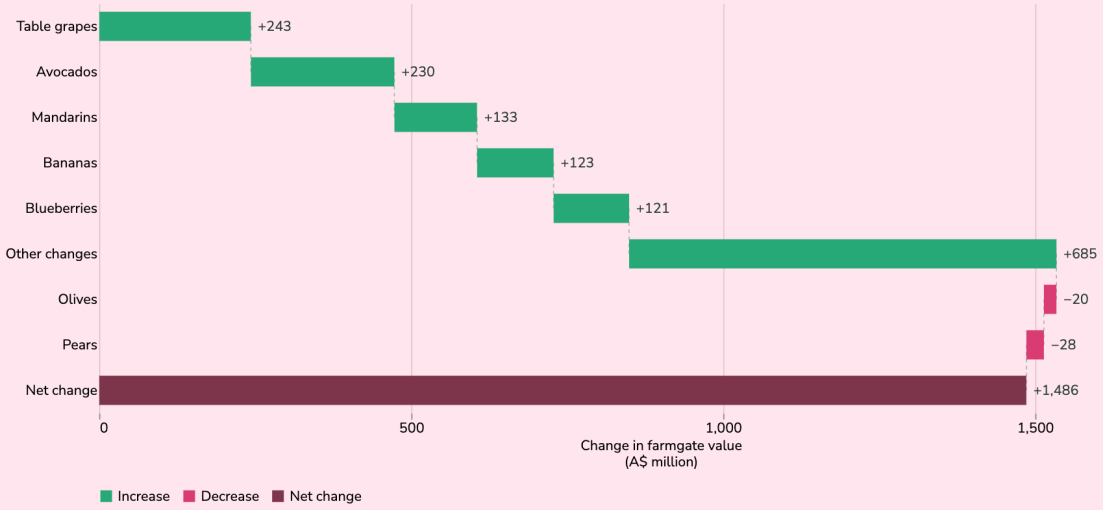
> **Apples:** production **rose 5.5%**; nominal farmgate value **rose 8.4%** between 2020–21 and 2024–25.

Lines follow financial-year order. Axes adjust to each fruit and do not start at zero. Source: ABS, Table 2, 2020–21 to 2024–25.

▶ Interpreting value

Where did the increase in value come from?

A **$1.486 billion increase** in reported fruit value, with table grapes and avocados the largest individual contributors, 2020–21 to 2024–25.



Five largest gains, two largest declines and the net change in the other 22 categories. Values are current dollars, not inflation-adjusted. Source: ABS, Table 2.

Beyond our borders

Australia exports fruit to markets across Asia, the Pacific and beyond. Different fruits have different overseas destinations, so this final map compares five export categories.

Where Australian fruit goes overseas

Eight largest destinations by export volume, calendar 2025. **Route colours match destinations in the table.**

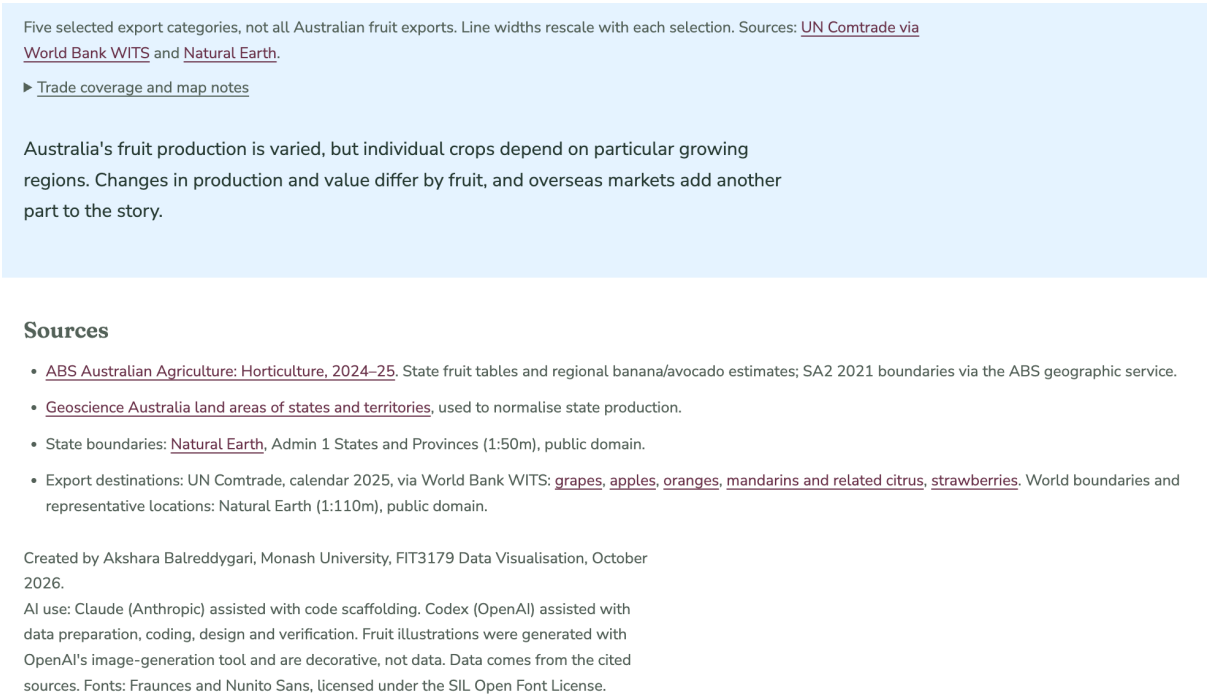


Destinations shown on the map

Destination	Tonnes	Share
China	113,084.6	24.3%
Japan	59,175.1	12.7%
Indonesia	47,641.7	10.2%
Hong Kong	37,951.8	8.2%
Thailand	25,822.2	5.5%
South Korea	25,031.8	5.4%
Vietnam	21,720	4.7%
Philippines	20,823.8	4.5%

Fruit: All five categories

The eight destinations shown account for 75.5% of 465,403.8 tonnes exported in this selection.



AI Declaration

For this assignment, I used Claude (Anthropic) and Codex (OpenAI) as support tools while developing my visualisation.

How I Used AI

I used AI as a sounding board for the story and layout, to help locate relevant public datasets, and to assist with preparing the data for the charts. It also helped with writing and revising code, troubleshooting chart behaviour, and checking calculations and layouts at different screen sizes.

I used AI to suggest improvements to headings, captions and annotations, and to help refine the assignment description. The decorative fruit illustrations were generated using OpenAI's image-generation tool. These are visual decorations only and do not represent data. All data comes from the sources cited in the visualisation.

I chose the topic and direction of the visualisation and guided the revisions through my own feedback and the feedback received from my tutor. I made the final decisions about which suggestions to use and how the charts, colours, layout and written explanations should fit together.

All content, code and design choices included in the submission have been reviewed and finalised by me. I take responsibility for the accuracy of the information, the chart choices and the final work submitted.